Available online at www.sciencedirect.com

ScienceDirect

Procedia Computer Science 165 (2019) 182–188

Procedia Computer Science

www.elsevier.com/locate/procedia

INTERNATIONAL CONFERENCE ON RECENT TRENDS IN ADVANCED COMPUTING 2019, ICRTAC 2019

Multiple Sub-Filter Adaptive Noise Canceller for Fetal ECG Extraction

Jaba Deva Krupa Abel[a], Dhanalakshmi Samiappan[a], R Kumar[a], Pravin Kumar S[b]

[a]*SRM Institute of Science and Technology, Kancheepuram 603203, India*
[b]*SSN college of Engineering, Chennai 603110, India*

Abstract

Non-invasive monitoring of fetal cardiac signs through the signal processing of maternal abdominal signals has been emerging as a promising technique in the field of obstetrics and gynaecology. The underlying challenges in obtaining the Fetal Electrocardiogram (FECG) from maternal abdominal signals include elimination of dominant maternal ECG and other noises. The simple technique based on adaptive noise cancellers uses Single Long Filter (SLF) as adaptive filter for extracting the FECG. In this paper, we propose an algorithm for implementation of adaptive filter in adaptive noise canceller (ANC) as Multiple Sub-Filters (MSF). Each of the adaptive sub-filters are updated using the Least Mean Square (LMS) algorithm. The proposed MSF architecture is compared with traditional SLF ANC by testing using a real data from Daisy database. From the obtained results, it is observed that MSF-ANC performs better than SLF-ANC with 91.66% positive predictive value and 84.61% accuracy.

Keywords: Fetal Electrocardiogram; Adaptive noise cancellers; Multiple sub-filter; Single long filter

1. Introduction

Traditional approach for monitoring the fetal health status is based on doppler ultrasound techniques, ultrasonography for antepartum and Cardiotocography (CTG) for intrapartum monitoring. These technique faces some potential issues including restricted continuous monitoring as it may impact the safety of fetus since it

* Corresponding author. Tel.: +918056256491
*E-mail address:*dhanalas@srmist.edu.in

10.1016/j.procs.2020.01.093

involves exposure to 2MHz ultrasound frequencies [1], risk of picking up the maternal heart rate instead of fetus [2], frequent repositioning of transducers and need of expertise staff in monitoring.

Fetal electrocardiogram can be the best alternative solution providing accurate heart rates and can be done through internal and external monitoring. The internal monitoring records direct electrocardiogram by placing an electrode on scalp after the rupture of membranes and dilation of cervix and is highly prone to infections. External monitoring is non-invasive technique which measures the fetal electrocardiogram from the electrophysiological signals obtained from mother's abdomen. Compared to its other counterparts, non-invasive fetal ECG is a safe technique providing continuous long-term monitoring and is receiving much research interests for the past decade.

The signal processing of non-invasive abdominal signals for obtaining fetal electrocardiogram was first demonstrated by Widrow [3], in which he proposed an adaptive noise canceller estimating the maternal component in the abdominal signal. Later several algorithms were addressed in the literature for the problem of fetal ECG extraction can be categorized as signal decomposition and adaptive filtering techniques. Signal decomposition approaches such as principal component analysis [4], independent component analysis [5], periodic component analysis [6] and sub-space decomposition [7] aims at suppression of mECG based on assumption that different source signals are linearly mixed which may not be true for a non-stationary environment like current problem of interest.

The adaptive filtering performs well in case of non-stationarity and several algorithms were presented using both linear and non-linear adaptive networks. Non-linear adaptive algorithms as proposed by Camps et al [8], Hasan et al [9] and Zhong et al [10] are all subjected to weight initialization problems. Linear adaptive techniques use adaptive noise cancellers for mECG suppression. As a modification to the technique proposed by Widrow [3], the ANC architecture with simple or multiple primary and reference channels, along with non-linear system like FLANN/GFLANN in parallel to FIR filter is proposed in ma et al [11] where the linear FIR filter is updated using LMS algorithm. The drawback with conventional LMS filter is its slow convergence while its counterpart Recursive least square (RLS) algorithm provides better convergence.

With a motivation that LMS algorithm is simple and less computationally complex compared to RLS algorithms and with an aim to improve the convergence of LMS algorithms, in this paper, we propose an adaptive noise canceller with multiple sub-band filter. Multiple sub filters are applied to the problem of acoustic echo cancellations [12] and the advantage of this structure is better convergence of Mean Square Error (MSE) compared to LMS in single long adaptive filters. We implement the multiple sub-filter adaptive noise canceller for suppressing the maternal ECG component in the acquired abdominal signals and compare the results with SLF in-terms of accuracy and sensitivity.

2. Related Work

In [18], the author proposed a method called Adaptive Neuro-fuzzy inference system (ANFIS) network to identify non-linearity between thoracic and maternal component in AECG. Followed by non- linearity mapping, FECG is obtained by direct subtraction of estimated MECG from the abdominal signal. This approach was tested using both real data available from a database and synthetic data. It is found that, this approach performs well even for complete fetal and maternal QRS complexes overlap and outperforms for fetal to maternal SNR below -20dB.

In [16], the authors proposed a non-linear Bayesian framework using Kalman filters for extracting the fetal ECG using single channel abdominal signal. This approach models the recorded signal as sum of Gaussian curves. FECG can be recursively extracted by considering one ECG at a time. Performance highly depends on correct detection of R-peaks and not satisfactory for overlapped scenarios.

Behar et al [17] proposed an algorithm using Echo State Neural network (ESN) based filtering for removing MECG. The ESN is a technique for training the recurrent neural network and it is used to update the adaptive filter. This technique proves to have high capability in approximating the non-linearity between maternal and thoracic signals but fetal R magnitudes are severely compressed. The drawback associated with this technique is higher computational load and suffers from weight initialization problem.

3. Multiple Sub-filter ANC architecture

The pioneering method used for denoising in non-stationary environment is the adaptive noise cancellers. The structure of adaptive noise canceller using a single filter is shown in figure 1. The filter continuously update its weight in order to adapt the changing scenarios and the udpation is based on minimization of the error signal e(n), which is difference between the signals m(n) and a(n). The cost function could either minimize mean square error or least square error based on which the adaptive algorithm for updating the filter coefficients will be Least Mean Square (LMS) or Recursive Least Square (RLS).

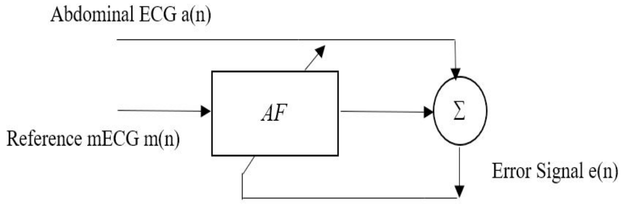

Fig. 1. Multiple sub-filter ANC structure for extraction of fetal ECG

The RLS algorithm involves complex computations compared to LMS still providing better convergence than LMS. The problem of low convergence in single long adaptive filters are addressed by partitioning the SLF of order N sinto K sub filters of lower orders. These multiple sub filters are parallelly updated using LMS algorithm reducing the load of single long filter and thereby improving the convergence. Figure 2 shows the structure of multiple sub-filter ANC having K sub-filters $\{AF_0, AF_1, \ldots AF_{K-1}\}$ with primary inputs $\{M_0, M_1, \ldots\ldots M_{K-1}\}$. The error signal for updating the filters is computed at the last stage as common error e(n) as shown in figure 1a. The single adaptive filter *AF* shown in figure 1 is realized as multiple parallel filters.

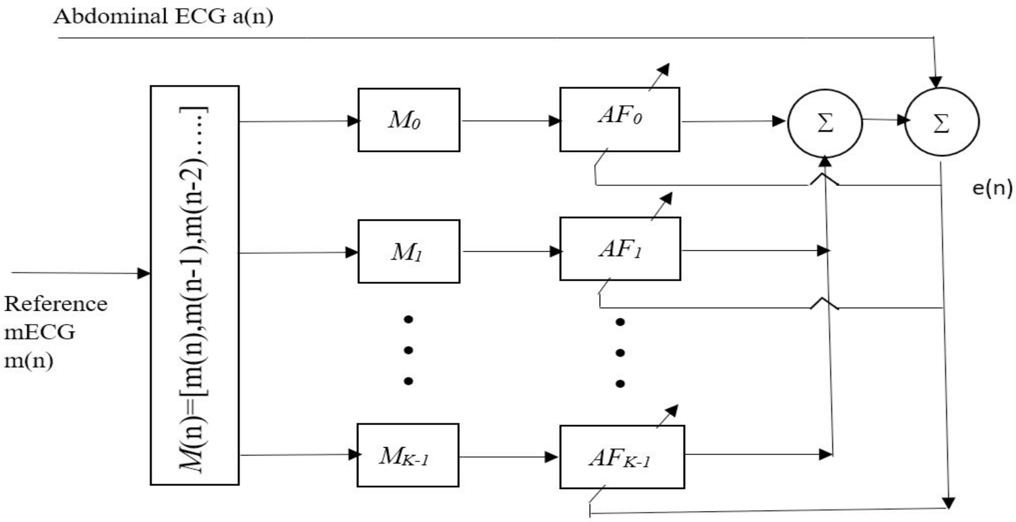

Fig. 1a. Multiple sub-filter ANC structure for extraction of fetal ECG

The adaptation algorithm for the proposed ANC structure is given below:

1. Input data:
   - The reference maternal ECG

$$M(n) = [m(n), m(n-1), ...]$$

- Input to j^{th} sub-filter is given by

$$M_j(n) = M[jL + n]; \qquad \forall j = 0,1,...K-1$$

Where L is length of each sub-filter.

2. Initialization: $W_j(0) = 0 \qquad \forall j = 0,1,...K-1$
3. For $n = 0,1,2........$
- Output of each sub-filter is given by

$$y_j(n) = M_j^T(n) W_j(n)$$

- Computation of error

$$e(n) = a(n) - \sum_{j=0}^{K-1} y_j(n)$$

- Updation of filter coefficients using LMS algorithm

$$W_j(n+1) = W_j(n) + \mu M_j(n) e(n) \qquad \forall j = 0,1,...K-1$$

The notations used in this article are listed in table 1.

Table 1. Notation Table

Notation	Description
ANC	Adaptive Noise Canceller
SLF	Single Long filter
MSF	Multiple Sub-Filter
FECG	Fetal Electrocardiogram
AECG - a(n)	Abdominal Electrocardiogram
MECG – m(n)	Maternal Electrocardiogram
AF	Adaptive filter
W(n)	Filter coefficients
μ	Step size
e(n)	Error signal
y(n)	Output of adaptive filter
L	Length of each sub filter
K	Number of sub filters

4. Experimental Results

The efficiency of proposed method has been examined using real data from Daisy Database [13]. This database consists of recordings from a single pregnant woman comprising a total of eight recordings lasting for duration of 10s and sampled at a frequency of 250Hz. The first five channels are the abdominal signals and the remaining three are the thoracic signals. The primary and reference channels of the ANC are taken to be the second abdominal and third thoracic signals of the Daisy database respectively. The pre-processing stage involves cascading of first order high pass filter with a cutoff frequency of 3Hz to remove low frequency noise such as electrohysterogram (EHG) whose spectrum ranges between 0.1Hz and 3Hz [14] and respiratory signals that cause baseline wander, and low pass Butterworth filter of cutoff frequency 65Hz to remove high-frequency noise components. A notch filter having a centre frequency of 50Hz is used to remove power line interference. The normalised pre-processed AECG signal and MECG signals considered as input to ANC is shown in figure 2.

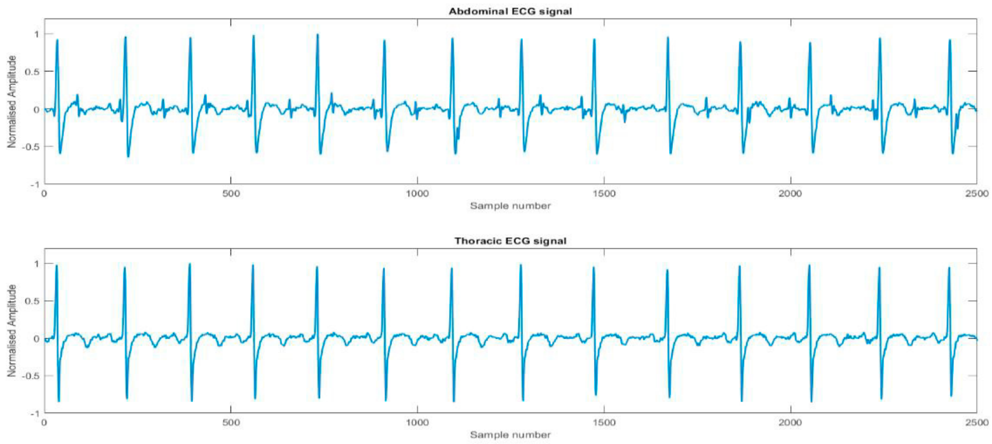

Fig. 2. Primary signal and reference signal to ANC

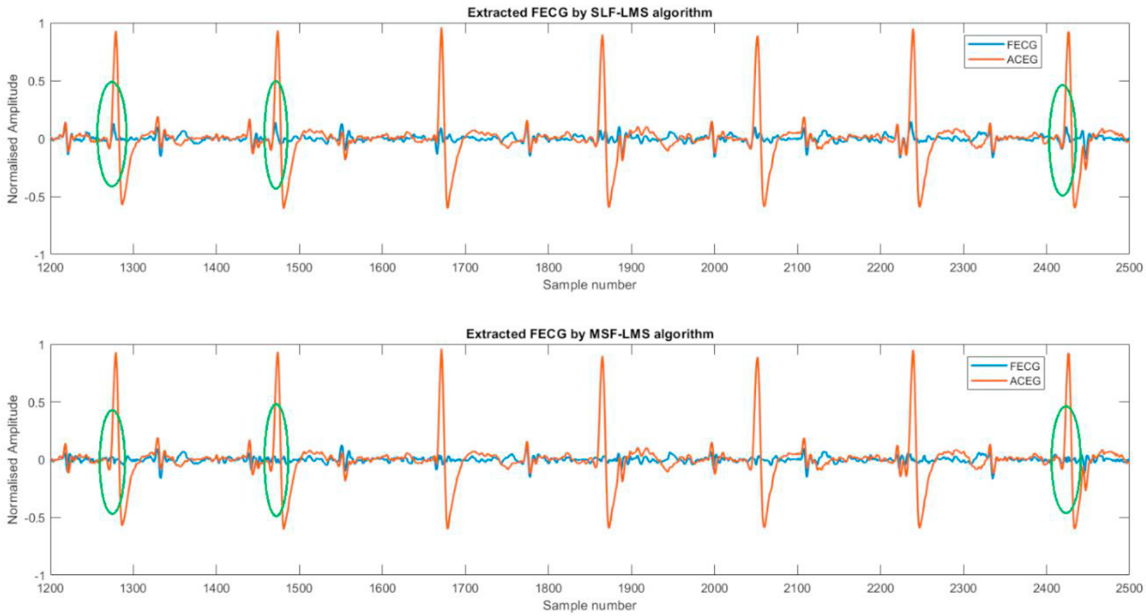

Fig. 3. Extracted fetal ECG using SLF-LMS and MSF-LMS ANC architectures

The input signals are fed to traditional single long filter (SLF-ANC) and the proposed multiple sub-filter (MSF-ANC). The proposed architecture is simulated using three sub-filters, each of same order and step size for simplicity. The extracted fetal ECG signal using SLF-ANC and MSF-ANC are depicted in figure 3. The plots show the output waveforms after the convergence of adaptive filters. From the obtained results it is observed that in the fetal ECG signal extracted using MSF-ANC, the maternal components are very well supressed than that of FECG extracted using SLF-ANC. In figure 3, it is observed that at sample 1276, 1470 and 2424 as highlighted, the maternal peaks are completely supressed using MSF-ANC than SLF-ANC and the reason for this is because of faster convergence of multiple sub-filter structure. The performance of the proposed adaptive noise canceller architecture for fetal ECG extraction can be quantitatively analysed by processing the obtained results using R-peak detector.

For the performance analysis of MSF-ANC, the extracted fetal ECG is processed using Pan-Tompkins [15] R-peak detection technique and the output obtained is shown in figure 4. Using the R-peaks detected, the evaluation metrics such as Sensitivity, Positive predictive value, Accuracy and F1 score are obtained and tabulated in Table 1.

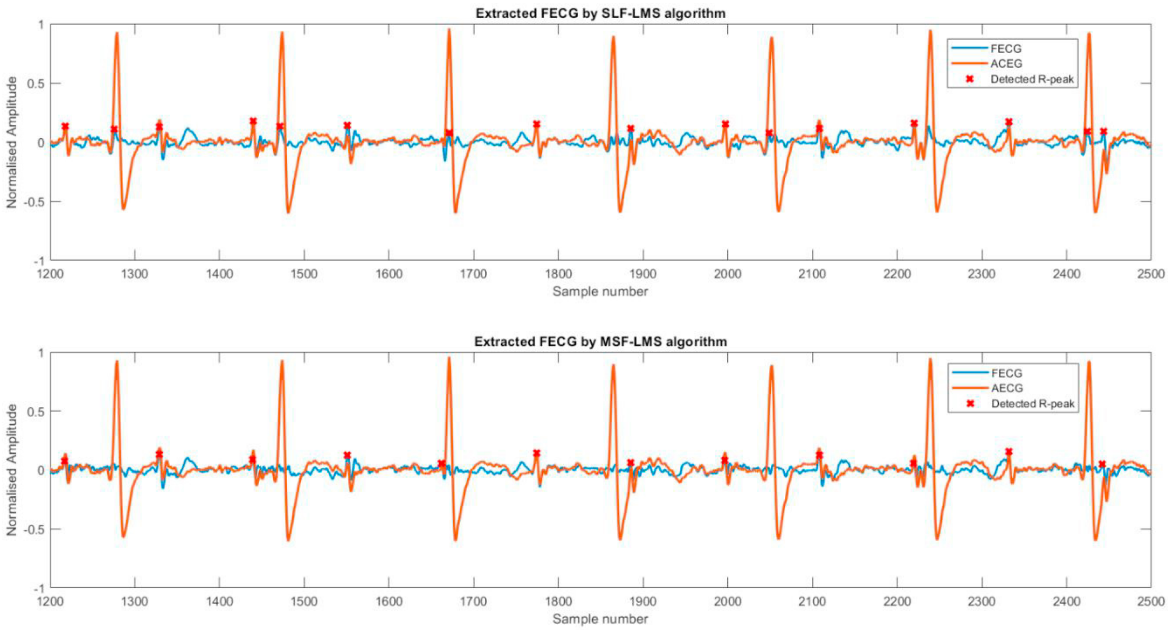

Fig. 4. Output of Pan-Tompkins R-peak detector

The performance metrics computed for multiple sub filter ANC and single long filter ANC as shown in table 1 reveals that multiple sub-filter ANC provides better accuracy, positive predictive value and F1-score compared to that of SLF-ANC. The quantitative measure of improvements in these factors will be 15.86%, 22.91% and 10.18% respectively. From these observations we can conclude that implementing the single adaptive filter as multiple sub filters yields significant improvement in ANC by providing 84.61% accuracy.

Table 2. Performance metrics computed for SLF and MSF ANC.

Parameters (%)	SLF-ANC	MSF-ANC
Sensitivity	100	91.66
Accuracy	68.75	**84.61**
Positive predictive value	68.75	**91.66**
F1 Score	81.48	**91.66**

5. Conclusion

In this paper, we have proposed a novel technique implementing the single adaptive filter as multiple sub-filters for extracting the fetal ECG from the abdominal signals using adaptive noise canceller. The performance of the proposed technique is tested using real data from Daisy database. Experimental results have demonstrated that the proposed approach behaves superior than traditional ANC by completely eliminating the maternal peaks in the extracted FECG. The quantitative performance comparison between SLF-ANC and MSF-ANC shows that proposed architecture improves the accuracy, positive predictive value and F1-score by 15.86%, 22.91% and 10.18% respectively. The future scope of work includes developing adaptive algorithm with variable step size which aims to achieve reduced mean square error.

References

[1] Barnett, S. B., and D. Maulik. "Guidelines and recommendations for safe use of Doppler ultrasound in perinatal applications." *Journal of Maternal-Fetal Medicine*10.2 (2001): 75-84.

[2] Neilson Jr, Duncan R., Roger K. Freeman, and Shelora Mangan. "Signal ambiguity resulting in unexpected outcome with external fetal heart rate monitoring." *American journal of obstetrics and gynecology* 198.6 (2008): 717-724

[3] Widrow, Bernard, et al. "Adaptive noise cancelling: Principles and applications." *Proceedings of the IEEE* 63.12 (1975): 1692-1716

[4] Kanjilal, Partha Pratim, Sarbani Palit, and Goutam Saha. "Fetal ECG extraction from single-channel maternal ECG using singular value decomposition." *IEEE Transactions on Biomedical Engineering* 44.1 (1997): 51-59.

[5] Hyvarinen, Aapo. "Fast and robust fixed-point algorithms for independent component analysis." *IEEE transactions on Neural Networks* 10.3 (1999): 626-634.

[6] Sameni, Reza, Christian Jutten, and Mohammad B. Shamsollahi. "Multichannel electrocardiogram decomposition using periodic component analysis." *IEEE transactions on biomedical engineering* 55.8 (2008): 1935-1940.

[7] Sameni, Reza, Christian Jutten, and Mohammad B. Shamsollahi. "A deflation procedure for subspace decomposition." *IEEE Transactions on Signal Processing* 58.4 (2009): 2363-2374.

[8] Camps, G., M. Martinez, and E. Soria. "Fetal ECG extraction using an FIR neural network." *Computers in Cardiology 2001. Vol. 28 (Cat. No. 01CH37287)*. IEEE, 2001.

[9] Hasan, Muhammad A., Muhammad I. Ibrahimy, and Mamun Bin Ibne Reaz. "Fetal ECG extraction from maternal abdominal ECG using neural network." *Journal of Software Engineering and Applications* 2.05 (2009): 330.

[10] Zhong, Wei, et al. "A deep learning approach for fetal QRS complex detection." *Physiological measurement* 39.4 (2018): 045004.

[11] Ma, Yaping, et al. "Foetal ECG extraction using non-linear adaptive noise canceller with multiple primary channels." *IET Signal Processing* 12.2 (2017): 219-227.

[12] Barik, Alaka, et al. "LMS adaptive multiple sub-filters based acoustic echo cancellation." *2010 International Conference on Computer and Communication Technology (ICCCT)*. IEEE, 2010.

[13] De Moor, B., De Gersem, P., De Schutter, B., & Favoreel, W. (1997). DAISY: A database for identification of systems. *JOURNAL A*, *38*, 4-5.

[14] Ionescu, Viorel. "Fetal ECG Extraction from Multichannel Abdominal ECG Recordings for Health Monitoring During Labor." *Procedia technology* 22 (2016): 682-689.

[15] Pan, Jiapu, and Willis J. Tompkins. "A real-time QRS detection algorithm." *IEEE Trans. Biomed. Eng* 32.3 (1985): 230-236.

[16] Niknazar, Mohammad, Bertrand Rivet, and Christian Jutten. "Fetal ECG extraction by extended state Kalman filtering based on single-channel recordings." IEEE Transactions on Biomedical Engineering 60.5 (2012): 1345-1352.

[17] Behar, Joachim, et al. "A comparison of single channel fetal ECG extraction methods." Annals of biomedical engineering 42.6 (2014): 1340-1353.

[18] Assaleh, Khaled. "Extraction of fetal electrocardiogram using adaptive neuro-fuzzy inference systems." IEEE Transactions on Biomedical Engineering 54.1 (2006): 59-68.